

Devkumar Banerjee

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EDUCATION

A&M Consolidated High School <i>High School Diploma</i> ▪ National Merit Semifinalist ▪ PSAT: 1520/1520 (Full Score) ▪ SAT: 1550/1600 (Math: 800/800, EBRW: 750/800) ▪ GPA (Weighted): 4.5082/4.00 ▪ GPA (Unweighted): 4.00/4.00 ▪ Rank: 19/506 ▪ Activities and societies: Orchestra (Concertmaster, received a scholarship to study privately), AI/ML-Cybersecurity Society (President), Texas Boys State Delegate, Mu Alpha Theta (Treasurer), Model UN (Treasurer), CyberPatriot Team, UIL (Computer Science, Mathematics, Number Sense, and Calculator), NHS (Member) ▪ AP Coursework: AP Calculus AB (5), AP Calculus BC (5), AP Statistics (5), AP Computer Science A (5), AP Physics 1 (4), AP Physics C: Mechanics (4), AP Physics C: Electricity and Magnetism (4), AP English Language and Composition (4), AP Spanish Language and Culture (3), AP United States History (4), AP World History (5), AP Human Geography (5), AP Chemistry, AP English Literature and Composition, AP Microeconomics, AP Macroeconomics, AP United States Government.	August 2021-Present <i>College Station, TX</i>
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Texas A&M University <i>Non-degree-seeking Mathematics</i> • Taking Calculus 3 (MATH 251) and Linear Algebra (MATH 304) as a high schooler at Texas A&M.	August 2024-Present <i>College Station, TX</i>
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WORK EXPERIENCE

Irma Rangel College of Pharmacy, Texas A&M University <i>Research Intern (36 wk/yr., 30 hrs/wk.)</i> ▪ Statistically analyzed publicly available transcriptomic data from NIH studies to confirm a link between phthalates, microplastic, Type 2 Diabetes Mellitus, and Obesity. ▪ Presented my research at the 2024 Texas Junior Academy of Sciences, placing 3rd.	May 2024-Present <i>College Station, TX</i>
UT Health Science Center - San Antonio <i>Research Intern (10 wk/yr., 60 hrs/wk.)</i> ▪ Participated in a paid, in-person research internship at UT Health Science Center - San Antonio ▪ Created a unified Pheochromocytoma and Paraganglioma (PCPG) data analysis and visualization platform for transcriptomics researchers, with plans to scale the project's scope to other types of cancers.	May 2024-August 2024 <i>San Antonio, TX</i>
American Legion Auxiliary <i>Boys State Delegate (10 wk/yr., 60 hrs/wk.)</i> ▪ Participated in one of the most respected and selective educational programs of government instruction for U.S. high school students. ▪ As a part of this opportunity, I participated and learned about roles in the local, county, and state governments.	March 2024-June 2024 <i>Austin, TX</i>

Clarkson University

Student Trainee (2 wk/yr., 10 hrs/wk.)

- Participated in an NSF-funded graduate-level training program as a high school junior.
- Worked with other graduate students from Stony Brook University and TU Dortmund using tools such as the Portable, Extensible Toolkit for Scientific Computation (PETSc) and the Scalable Library for Eigenvalue Problem Computations (SLEPc) to solve eigenvalue problems and Ordinary Differential Equations (ODEs).
- Simulated Proper Orthogonal Decomposition (POD), reducing the computational intensity of scientific modeling tasks.

January 2024

Potsdam, NY (Remote)

Houston Methodist Research Institute

Machine Learning in Nanomedicine Intern (10 wk/yr., 60 hrs/wk.)

- In-person paid research internship at the Houston Methodist Research Institute in the Texas Medical Center
- Utilized statistical analysis tools to identify important physiochemical and experimental characteristics of in vivo subjects and injected nanoparticles (NPs) to help shed light on the most important factors in NP Tumor Biodistribution.
- Participated in a panel discussion alongside Texas A&M Univ. Engineering Medicine (EnMed) students, offering guidance to high school students exploring their medical career interests.
- Received the Waggoner Foundation Merit Scholarship for my work.
- Presented my work at the Houston Methodist Summer Science Symposium 2023.
- Received 30+ Bariatric Surgery Shadowing Hours

March 2023-Present

College Station, TX

Kumon North America, Inc.

Mathematics Tutor (52 wk/yr., 3 hrs/wk.)

- Created a personalized environment for students to maximize their potential for achievement in mathematics (STEM).
- Emphasized critical thinking and independent thought among students.

August 2021-Present

College Station, TX

Self Employed

Mathematics/Science Tutor (15 wk/yr., 6 hrs/wk.)

- Created personalized lesson plans for 10 students to maximize their potential for achievement in mathematics and computer science by emphasizing critical thinking and independent thought, thus allowing said students to learn math above their grade level and build their own personal computers.

June 2022-Present

College Station, TX

VOLUNTEERING EXPERIENCE

Texas A&M University

Junior Counselor (10 wk/yr., 8 hrs/wk.)

- Mentored gifted students (in the 6th, 7th, and 8th grades) to promote STEM education nationally in diverse communities (especially minority students) at the SEE-MATH camp (an honors mathematics camp for gifted students).
- Trained SEE-MATH campers to develop digital animations using the Maple Computer Algebra System (CAS).
- URL: <https://see-math.math.tamu.edu/>

July 2020 – Present

College Station, TX

Texas A&M University Present

Junior Counselor (10 wk/yr., 8 hrs/wk.)

- Mentored gifted students (in the 6th, 7th, and 8th grades) to promote cybersecurity-related topics in diverse communities (especially minority students) at the TAMU HPRC's annual

June 2020 –

College Station, TX

Summer Computing Academy (SCA).

- Mentored campers in various computing topics, including Machine Learning and Python Programming.
- Interviewed by a local TV station about the camp, inspiring and recruiting younger students to apply for this camp (Timestamp: 1:05): <https://bit.ly/CampSmartInterview2022>

Aggieland Country School

Instructor (2 wk/yr., 20 hrs/wk.)

- In 9th grade, I created and implemented a curriculum introducing elementary-school students (Grades 1-4) to Computer Science and programming using Scratch and Scratch Jr.

June 2022

College Station, TX

Safe Tech Inc. (STI)

Instructor (30 wk/yr., 10 hrs/wk.)

- In the 10th grade, I designed and programmed an I²C driver for the Raspberry Pi Sense HAT for use with an ESP32 Microcontroller.
- As a part of the “Project Lead-The-Way (PLTW),”: I worked with Mickie Byrd (President and founder of STI) and trained STEM teachers from Brownsville ISD to perform Python programming (using Jupyter Notebooks) for use in their curriculum in the AggieSTEM Summer Program (“train-the-trainer” for minority-serving institutions).
- Designed, programmed, and implemented autonomous vehicles (rovers) using computer vision (CV).

June 2022-Present

College Station, TX

Brazos Valley Food Bank

Volunteer (10 hours)

- Helped sort items and arranged food boxes for delivery to local food pantries.

December 2022

College Station, TX

AWARDS AND DISTINCTIONS

3rd place - Texas Junior Academy of Sciences (TJAS)

October 2024

Texas A&M University

I conducted research throughout the summer on confirming a link between phthalates and microplastics with Type 2 Diabetes Mellitus and Obesity. I wrote a 20-page paper and a 12-minute presentation, winning 3rd place out of hundreds of applicants (Top 0.6%).

Boys State Delegate

June 2024

American Legion - Texas Boys State

I gained hands-on experience in legislative, executive, and judicial processes and developed critical thinking, public speaking, and teamwork skills through active participation in an intensive week-long program simulating state government operations. I engaged in leadership workshops focused on civic engagement and ethical decision-making. I was elected as one of the top 1000 students in Texas (Top 0.23%) for this prestigious program

5th Place - Information Technology Fundamentals Open Event

May 2024

Business Professionals of America - National Leadership Conference

I placed 5th out of 40000 (Top 0.0125%) BPA competitors by showcasing my expertise in information technology concepts, programming language evolution and their applications, hardware and software basics, graphics concepts, printers, operating systems, security, law, and ethics, ergonomics, and environmental concern, as well as Network concepts.

3rd Place - Linux Operating System Fundamentals

May 2024

Business Professionals of America - National Leadership Conference

I placed 3rd out of 40000 (Top 0.0075%) BPA competitors by showcasing my expertise in networking, filesystem management, process management, and security hardening techniques. My performance demonstrated a comprehensive understanding of the GNU/Linux kernel and operating system fundamentals of common Linux Distributions (including Ubuntu, Fedora, and

Red Hat Enterprise Linux).

2024 National Cyber Scholar with Honors

February 2024

National Cyber Scholarship Foundation

I was recognized as a National Cyber Scholar with Honors by completing cybersecurity challenges on the CyberStart Platform.

Badge: <https://api.badgr.io/public/assertions/t6xS3chbQACURHaJMbQ7kw>

Best Beginner Hardware Hack

January 2024

TAMUHack

Placed 4th out of more than 1000 (Top 0.04%) people for creating a Convolutional Neural Network and microcontroller-based voice authentication module.

Repository: <https://github.com/aptitudepi/2024TAMUHack>

Top 100 Overall - National Semifinalists

December 2023

Air Force Association - CyberPatriot

In the 11th grade, I was ranked in the top 150 (Top 5%) nationally in the Cyberpatriot competition for securing various computers running the Windows and Linux Operating Systems.

2023 National Cyber Scholar with Honors

October 2023

National Cyber Scholarship Foundation

I was recognized as a National Cyber Scholar with Honors by completing cybersecurity challenges on the CyberStart Platform.

Badge: <https://api.badgr.io/public/assertions/t6xS3chbQACURHaJMbQ7kw>

3rd Place - Hospital Plunge

October 2023

TAMU Datathon

Placed 3rd out of more than 600 (Top 0.5%) people for creating a Convolutional Neural Network to predict healthcare outcomes.

AP Scholar with Distinction

July 2023

The College Board

I have earned the AP® Scholar with Distinction Award for achieving an average score of at least 3.5 on all AP Exams taken, and scores of 3 or higher on five or more exams.

2023 George H.W. Bush Essay Contest - Honorable Mention

April 2023

George H.W. Bush Presidential Library & Museum

I received the "Honorable Mention" prize (i.e., 4th place) out of more than 100 (Top 4%) applicants for writing an essay on cultural exchange.

Top 100 Overall - National Semifinalists

December 2022

Air Force Association - CyberPatriot

In the 10th grade, I was ranked in the top 100 (Top 3.33%) nationally in the Cyberpatriot competition for securing various computers running the Windows and Linux Operating Systems.

1st Prize Overall

July 2022

Texas A&M University - Governor's Summer Merit Academy

Won the 1st prize overall in the Texas A&M University HPRC Summer Computing Academy 2022 for developing a Machine Learning (ML)-based model for diagnosing diseases based on an input of two distinct symptoms

All-Region (1st Violin) - 3rd Chair

October 2021

Texas Music Education Association (TMEA) Region 8

Selected as the third chair, first violinist of the TMEA Region 8 Orchestra.

5th Prize Overall - as a Mentor

July 2021

Texas A&M University (TAMU) - Governor's Summer Merit Academy

In my capacity as a middle-school student, I mentored a team of high-school students (for their team project on analyzing the effect of tweets on cryptocurrency prices). The team won the 5th prize overall in the HPRC Summer Computing Academy (SCA 2021) at Texas A&M University, College Station, TX.

3rd Prize Overall

June 2021

Texas A&M University (TAMU) - Governor's Summer Merit Academy

I led my team to win the 3rd prize in the TAMU HPRC Summer Computing Academy for developing a cybersecurity model for use in hospital networks to prevent hacking of patient data and networks.

1st Place - Hall of Fame

June 2019

NSA/NSF Gencyber Program

Placed 1st in the Gencyber Program Hall of Fame for identifying security flaws on the platform used for the joint NSA- and NSF-led GenCyber program.

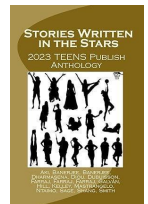
PUBLICATIONS & PRESENTATIONS (CONFERENCE/ WORKSHOP)

Stories Written in the Stars (Book/ Anthology) Story (Novel, Fiction): Terminus

21st November 2023

Amazon KDP (<https://bit.ly/StoriesInTheStars2023>)

Get ready to embark on a miraculous literary journey. Within the pages of this captivating collection, you'll encounter relatable characters, partake in exhilarating adventures, and delve into imaginative worlds where anything is possible. These remarkable stories, spanning from poetry to short fiction to novel excerpts, were meticulously crafted over eight weeks. With diverse genres including fantasy, horror, contemporary, dystopian, and thriller, this TEENS Publish anthology showcases the brilliant minds of seventeen creative individuals, each offering a unique perspective on the ever-changing adventure of imagination.



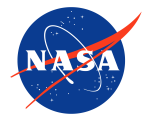
Benchmarking Performance of Machine Learning Algorithms for Fluid Flow Simulation by Leveraging Cluster Computing: A Case Study for High-Performance Computing

7th September 2022

NASA Thermal and Fluids Analysis Workshop 2022 (ONLINE)

Presented my research on benchmarking the performance of AI/ML algorithms by leveraging cluster computing capabilities on multiple Raspberry Pi computers using SLURM (Simple Linux Utility for Resource Management) for IoT (Internet of Things) applications. Presentation slides and recording are available at:

<https://go.nasa.gov/3VABkhv>; <https://youtu.be/famiRrm6Auk>

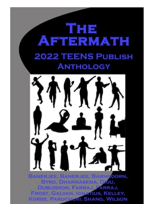


The Aftermath (Book/ Anthology) Story (Novel, Fiction): Artificial

1st November 2022

Amazon KDP (<https://bit.ly/TheAftermath2022>)

17 creative minds. 3000 words. 8 weeks. From poetry to short stories to novel excerpts, 17 imaginative authors write about difficult struggles in the world. Through these pages, unforgettable characters share their stories of imagination and courage. stories of fantasy and history to contemporary and thriller, we have stories for all. prepare for thrilling, heart-stopping tales.

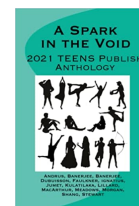


A Spark In The Void (Book/Anthology) Story (Novel, Fiction): Anomaly

24th November 2021

Amazon KDP (<https://bit.ly/SparkInTheVoid>)

Despite the hardships, separation, and isolation from each other and the rest of the world, fourteen brave young authors have virtually united to light a spark in the void of summer 2021. Three thousand words. Novel excerpts and short stories ranging from murder mysteries and rebellions to Earth-shattering catastrophes. Page after page, unforgettable characters and their daring, awe-inspiring stories that lie in these pages will draw you in and weave threads of creativity. Prepare for a little heat in this cold world.

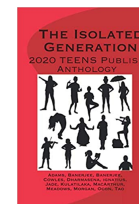


The Isolated Generation (Book/Anthology) Story (Novel, Fiction): The Mind

9th November 2020

Amazon KDP (<https://bit.ly/IsolatedGeneration>)

Thirteen incredible young writers unite to tell breathtaking stories, despite their isolation from each other and the rest of the world. On every page are awe-inspiring stories of danger, love, struggles, and the successes of many unforgettable characters. Read and discover where these authors go when they cannot leave their homes and the magic, suspense, and wonder they left for you to find.



MY Math Apps Calculus

9th November 2020

MY Math Apps (<https://mymathapps.com/mymacalc-sample/Credits.html>)

I have assisted Dr. Phil Yasskin (Professor, Math Dept., Texas A&M) in formulating/ coding, testing & publishing online exercises / interactive problems & plots/animations using JavaScript, Maple, and Sage for College Calculus 1 & 2 in the **MY-MATH-APPS** textbook (<https://bit.ly/myma-calc-2023-db>).



This textbook is taught to college students at Texas A&M & Univ. of South Carolina.

I created the pages on Integral Calculus: <https://bit.ly/myma-calc-area-2023-db>, Chain Rule <https://bit.ly/myma-calc-online-chain-tutorial-2023-db>, and Differential Calculus: <https://bit.ly/myma-calc-tangent-line-2023-db>

PROJECTS

Name	Description	URL(s)	Date
Mole Day Calculator	A Streamlit site designed to convert to/from moles, with integrations for all periodic table elements and possible compounds. Written for (and on) Mole Day (10/23/2024).	https://github.com/aptitudepi/mole-day ; https://mole-day.streamlit.app	2024/10/23
TIDALHack	A full-stack site using Groq's llama models to convert user input into input for an XGBoost classifier that classifies any of the 130+ symptoms into 40+ diseases.	https://github.com/aptitudepi/TIDALHack ; https://doctor-robot.streamlit.app/	2024/10/20
US Cyber Games CTF Challenge	A Capture-the-Flag style challenge where one takes advantage of an SQL	https://github.com/aptitudepi/USCG-SIV-GroupProject	2024/09/01

	Injection vulnerability in a hacker forum built on the LAMP (Linux/Apache/MySQL/PHP) stack.		
PCPG Analyzer	A Streamlit/web-based framework to dynamically analyze Pheochromocytoma and Paraganglioma (PCPG) transcriptome data to assist underfunded researchers from across the globe.	https://github.com/aptitudepi/pcpg-analyzer ; https://pcpg-analyzer.streamlit.app/	2024/08/13
Java Chess	A game of chess built entirely using Java Swing. Take the king to win!	https://github.com/aptitudepi/UnChackChess	2024/03/23
LinuxMAN	A compiled 2,537 PDF E-Book of manual and reference pages on the Linux Operating System (useful for Linux Operating System Fundamentals for Business Professionals of America)	https://github.com/aptitudepi/LinuxMAN	2024/03/09
TAMUHack	An AI-enabled voice authentication MFA solution.	https://github.com/aptitudepi/2024TAMUHack	2024/01/28
TAMU Datathon	A machine learning model to predict a patient's hospital length of stay.	https://github.com/aptitudepi/Hospital-Plunge	2023/10/29
ESP32 Driver	An I2C driver built as an interface between the HTS221 humidity sensor (on the Raspberry Pi SenseHAT) and the ESP32 microcontroller	https://github.com/aptitudepi/esp32_hts221_i2c_driver	2022/09/14

CERTIFICATIONS, LICENSES, AND CREDENTIALS (ABRIDGED)

For a full list of all 150+ certifications, licenses, and credentials, please see: <https://bit.ly/BanerjeeDevkumarFullCertList>

Authority	Name	URL	Date
Microsoft	Microsoft Certified: Azure AI Fundamentals	credly.com/badges/0a2d15ce-9227-4568-88b2-96c71e769eea	5/1/2024

PMI	PMI Project Management Ready	credly.com/badges/b05dd4a5-2201-4e6a-ba56-bc531a6c89d7	5/1/2024
Cisco	Cisco Certified Support Technician Cybersecurity (CCST Cybersecurity)	credly.com/badges/049a56de-fb70-40cf-b1a6-73ef1cfc309f	5/1/2024
Cisco	Cisco Certified Support Technician Networking (CCST Networking)	credly.com/badges/d2e41865-937d-448a-bb38-1668136d5e40	5/1/2024
Microsoft	Microsoft Certified: Security, Compliance, and Identity Fundamentals	credly.com/badges/840cb7d6-a874-4ac9-ac28-bf1219114277	5/1/2024
Microsoft	Microsoft Certified: Azure Fundamentals	credly.com/badges/0f9675bf-a575-4990-8efd-1844ba9b1593	5/1/2024
Certiport	Master Certification in Entrepreneurship	credly.com/badges/c0aa877b-ec6b-4a53-8294-591ee5babddf	5/1/2024
GIAC	GIAC Security Essentials Certification (GSEC)	credly.com/badges/e2e642f7-fd5d-43ac-a162-1fae3feba20f	5/1/2024
TestOut	TestOut Linux Pro	certification.testout.com/verifycert/6-1C6-VG5V7R	4/1/2024
GIAC	GIAC Foundational Cybersecurity Technologies (GFACT)	credly.com/badges/66c3c457-5bf5-412a-889b-061211f3fd16	8/1/2023

SKILLS & INTERESTS

Skills: 3D Printing, C++, Cascading Style Sheets (CSS), Computer-Aided-Design (CAD), Docker, HTML, ESP32 Microcontroller, FASTA (Data Format), Git, Javascript (JS), jQuery, Linux, Machine Learning, Microsoft Office (Excel, PowerPoint, and Word, Node.js, Pandas (Statistical Software), Python (Programming Language), PyTorch, PyTorch Lightning, Sage, SASS, SLURM (Simple Linux Utility for Resource Management), Tensorflow, Transact-SQL.

Interests: Computer Engineering/ Science (Programming), Data Science (AI), Mathematics, Medicine, 3D Printing

LANGUAGES

▪ Languages:

- o Bangla - Native proficiency
- o Bengali - Native proficiency
- o English - Native proficiency
- o Spanish - Working Proficiency
- o Hindi - Limited Working Proficiency